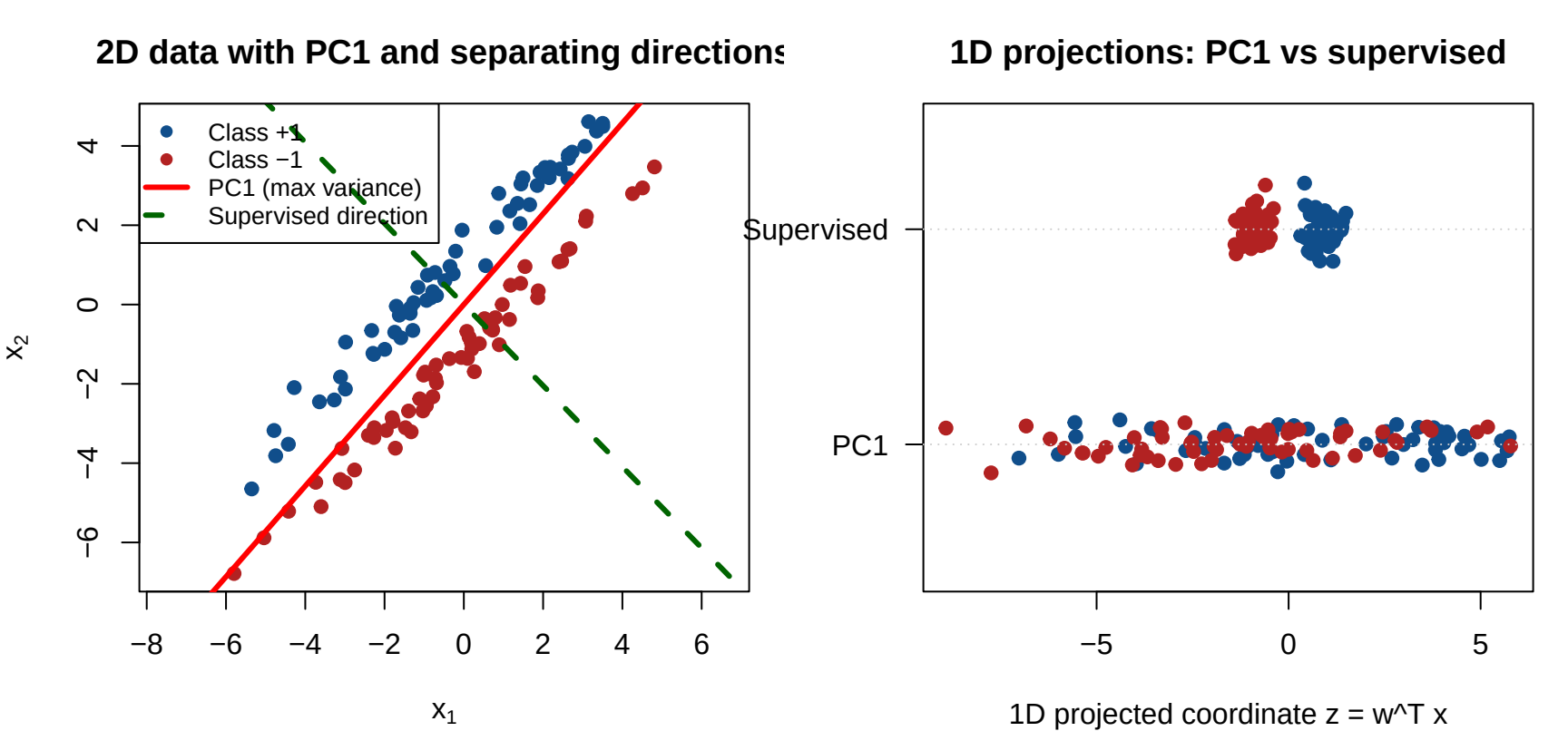
2D data with PC1 and separating directions
Class +1
Class −1
PC1 (max variance)
Supervised direction
x2
x1
−8
−6
−4
−2
0
2
4
6
−6
−4
−2
0
2
4
1D projections: PC1 vs supervised
Supervised
PC1
−5
0
5
1D projected coordinate z = w^T x